

Sunrise Inc.

PETALERT

TEST PLAN

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Introduction

The Test Plan has been created to communicate the test approach to team members. It includes the objectives, scope, schedule, risks and approach. This document will clearly identify what the test deliverables will be and what is deemed in and out of scope.

1.1 Objectives

- 1.2 Validate that pet profiles can be created, edited, deleted, and saved in Firebase Firestore.
- 1.3 Confirm that photo uploads to Firebase Storage persist after app restarts and across devices.
- 1.4 Verify that microchip numbers are saved and retrieved correctly from Firestore.
- 1.5 Confirm that reminder notifications trigger based on the configured schedule.
- 1.6 Test that the “Generate Alert” workflow produces correct email content and attachments using cloud-stored data.
- 1.7 Evaluate usability, color contrast, and navigation flow according to accessibility goals.

1.8 Team Members

Resource Name	Role
Tanvi Biswal	Developer
Kimberley Juarez	Developer
Anjali Fernando	Developer
Bibek Pandey	Developer
Ojaswi Subedi	Developer

2 Scope

The initial sprint will include ‘must have’ requirements. These and any other requirements that get included must all be tested.

The following sections indicate what is tested during each sprint. The scope of testing is determined at the beginning of the current sprint.

At the end of Sprint 1, a user must be able to:

1. Read the name of the app, PetFinder

At the end of Sprint 2, a user must be able to:

1. Add and view a profile for their pet
 - a. Testing will be conducted to ensure new profiles save and display information correctly
2. View the home page of the app

At the end of Sprint 3, a user must be able to:

1. Log in as an existing user or sign up as a new user to the app

- a. Security testing will be conducted to ensure logins work correctly and as intended
2. View a simple infographic of toxic foods, so that a user can quickly learn what to avoid
3. Use an alert system to send out information about a missing pet
 - a. Testing will be conducted to ensure information that is being sent out is accurate and up-to-date according to the last user imputed data
4. Upload more than one picture of his or her pet onto the app's photo gallery for a pet
 - a. Testing will be conducted to ensure that the pictures stay saved and display correctly on the app
5. View a recommendations section that displays treats or toys a listed pet in the user's pet profiles might enjoy
6. Change the schedule or timing in which they receive notifications to update pet information
 - a. Testing will be conducted to ensure that changes made to the scheduled notification times are correct and notify the user during the time they set
 - b. Testing will be conducted to ensure that the notifications work

At the end of Sprint 4, a user must be able to:

1. Create, edit, and delete pet profiles stored in Firebase Firestore
 - a. Testing will ensure data persists after app restart
 - b. Testing will confirm data synchronization across devices
2. Upload and retrieve pet images from Firebase Storage
 - a. Testing will verify image persistence across sessions
3. Add and view microchip numbers within pet profiles
 - a. Testing will ensure microchip data is saved and displayed correctly
4. Access pet data only after successful authentication
 - a. Security testing will confirm protected data access

Assumptions / Risks

2.1 Assumptions

We are working under the assumption that a user is someone who has access to a personal smartphone with the capabilities to download PetFinder, take pictures with his or her smartphone, and input information into required spaces on the app. We are also working under the assumption that a user will be able to modify and save changes he or she has made within the app, whether that be in the scheduled notification timings, pet profiles, emergency contacts, or alert system. We assume users have internet connectivity for cloud-based data synchronization through Firebase services.

2.2 Risks

The following risks have been identified, and the appropriate action has been identified to mitigate their impact on the project. The impact (or severity) of the risk is based on how the project would be affected if the risk was triggered. The trigger is what milestone or event would cause the risk to become an issue to be dealt with.

#	Risk	Impact	Trigger	Mitigation Plan
1	Scope Creep – as testers become more familiar with the tool, they will want more functionality	High	Delays in the implementation date	Each iteration, functionality will be closely monitored. Priorities will be set and discussed by stakeholders. Since the driver is functionality and not time, it may be necessary to push the date out.
2	Weekly delivery is not possible because the developer works off site	Medium	Product did not get delivered on schedule	
3	Firebase service outage	High	Cloud service unavailable	Implement proper error handling and retry logic
4				
5				

3 Test Approach

The project is using an agile approach, with 3-week sprints. *Mention how you will conduct testing during the sprint in terms of the techniques you plan to do and when. Add a new subsection for each sprint.*

Sprint 1

No testing because no implementation has been completed

Sprint 2

Testing will be conducted during the implementation of a feature and after the implementation completion of a feature to see if the feature fully adheres to what was

stipulated in the scope. Test reviews will be discussed at every meeting the team holds, and the test cases will be changed if deemed necessary.

Sprint 3

Testing will be conducted in the same manner as sprint 2 and test reviews will also function the same way and be held after team meetings. This can change in the future.

Sprint 4

Testing will focus on verifying Firestore integration, image storage using Firebase Storage, cross-device synchronization, and validation of the new microchip number field.

Cloud-based persistence and authenticated access will be prioritized. Testing will include regression testing of previously implemented features to ensure backend migration did not break existing functionality.

3.1 Test Automation

To reduce time in testing, the team plans on having a small database testers can pull information from to help ease testing. We will also use AI to create test cases for our project to save time.

3.2 Test Cases (Black Box)

3.2.1 Feature 1 - Pet Profiles

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-201	Create pet profile in Firestore	R1 – Pet Profile	Log in to the app. Navigate to Pets → Add Pet. Enter pet name, species/breed, and other required fields. (Optional) Enter microchip number. Tap Save.	A new pet profile is created and displayed in the pet list, and the pet data is stored in Firebase Firestore. The pet remains available after app restart.
TC-202	Edit pet profile and verify Firestore update	R1 – Pet Profile	Log in to the app. Open an existing pet profile. Tap Edit. Change one or more fields (ex, name, notes, breed, chip number). Tap Save.	The updated pet information is displayed correctly and persists after app restart. Firestore reflects the updated values.

Test Case ID	Description	Requirements Trace	Directions	Expected Output
			Close and reopen the app, then open the same pet profile again.	
TC-203	Delete pet profile from Firestore	R1 – Pet Profile	Log in to the app. Open pet list. Select a pet profile. Tap Delete and confirm deletion. Refresh pet list or restart app.	The pet profile is removed from the pet list and deleted from Firebase Firestore. The deleted pet does not reappear after restart.
TC-204	Save and retrieve microchip number	R1 – Pet Profile	Log in to the app. Add a new pet OR open an existing pet and tap Edit. Enter a microchip number in the chip field. Tap Save. Reopen the pet profile (or restart app and reopen).	The microchip number is saved successfully and displayed correctly in the pet profile. The value persists after restart and matches what was entered.
TC-205	Upload image to Firebase Storage	R1 – Pet Profile	Log in to the app. Add a new pet OR edit an existing pet. Tap Add Photo (camera or gallery). Select/take a photo and confirm. Tap Save. Restart app and reopen pet profile.	The pet image uploads successfully to Firebase Storage and displays correctly in the profile. The image persists after restart and loads correctly from cloud storage.

3.2.2 Feature 2 - Alert System

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-101	Prepare to request a reservation at a restaurant.	RES-002 RES-003 RES-005 RES-006	1. Execute TCB-004.	All fields necessary to fill in the data specified in the directions are present.

3.2.3 Feature 3 - Notification System

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-101	Prepare to request a reservation at a restaurant.	RES-002 RES-003 RES-005 RES-006	2. Execute TCB-004.	All fields necessary to fill in the data specified in the directions are present.

3.2.4 Feature 4 - Photo Gallery

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-101	Prepare to request a reservation at a restaurant.	RES-002 RES-003 RES-005 RES-006	3. Execute TCB-004.	All fields necessary to fill in the data specified in the directions are present.

3.2.5 Feature 5 - Recommendations Section

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-101	Prepare to request a reservation at a restaurant.	RES-002 RES-003 RES-005 RES-006	4. Execute TCB-004.	All fields necessary to fill in the data specified in the directions are present.

3.2.6 Feature 6 - Toxic Foods Infographic

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-101	Prepare to request a reservation at a restaurant.	RES-002 RES-003 RES-005 RES-006	5. Execute TCB-004.	All fields necessary to fill in the data specified in the directions are present.

3.3 Test Cases (White Box)

3.3.1 Feature 1 - Pet Profiles

Test Case ID	Description	Directions/Goals	Expected Output
TC-2001	Verify Firestore document structure matches model	Verify that the Firestore document fields match the expected pet model fields (ex: name, species, breed, age, notes, chipNumber, imageUrl, createdAt/updatedAt if used). Confirm required fields exist and field types are correct.	Firestore documents contain the correct field names and data types. No missing required fields. App can read/write without mapping errors.
TC-2002	Verify asynchronous data retrieval does not cause null pointer errors	Load the pet list and pet detail screens multiple times under different conditions (fresh login, slow network, app restart). Watch logs/console for null pointer exceptions or build/render errors during async fetch.	No crashes or null pointer exceptions. UI shows loading state while fetching, then displays data correctly once retrieved.

3.3.2 Feature 2 - Alert System

Test Case ID	Description	Directions/Goals	Expected Output
TC-1001	Selecting a restaurant to visit.	Follow TCB-001 when there are no restaurants available. Does the system handle not having a restaurant to select?	Should load the home page without errors. However, there are no database checks for null pointers so if the DB fails, there will be an error.

3.3.3 Feature 3 - Notification System

Test Case ID	Description	Directions/Goals	Expected Output
TC-1001	Selecting a restaurant to visit.	Follow TCB-001 when there are no restaurants available. Does the system handle not having a restaurant to select?	Should load the home page without errors. However, there are no database checks for null pointers so if the DB fails, there will be an error.

3.3.4 Feature 4 - Photo Gallery

Test Case ID	Description	Directions/Goals	Expected Output
TC-1001	Selecting a restaurant to visit.	Follow TCB-001 when there are no restaurants available. Does the system handle not having a restaurant to select?	Should load the home page without errors. However, there are no database checks for null pointers so if the DB fails, there will be an error.

3.3.5 Feature 5 - Recommendations Section

Test Case ID	Description	Directions/Goals	Expected Output
TC-1001	Selecting a restaurant to visit.	Follow TCB-001 when there are no restaurants available. Does the system handle not having a restaurant to select?	Should load the home page without errors. However, there are no database checks for null pointers so if the DB fails, there will be an error.

3.3.6 Feature 6 - Toxic Foods Infographic

Test Case ID	Description	Directions/Goals	Expected Output
TC-1001	Selecting a restaurant to visit.	Follow TCB-001 when there are no restaurants available. Does the system handle not having a restaurant to select?	Should load the home page without errors. However, there are no database checks for null pointers so if the DB fails, there will be an error.

Test Environment

Testers will create a new user using Firebase Authentication and test data persistence using Firebase Firestore and Firebase Storage. Testing will be conducted on Android emulator, physical devices, and across multiple authenticated sessions to verify cloud synchronization.

4 Test Schedule

Task Name (<i>sample is below, focus on spring 1 to start</i>)	Start	Finish	Effort	Comments
<i>Test Planning</i>				
<i>Review Requirements documents</i>				
<i>Create initial test estimates</i>				
<i>Learn new test resources</i>				
<i>First deploy to QA test environment</i>				
<i>Functional testing – Sprint 1</i>				
<i>Iteration 2 deploy to QA test environment</i>				
<i>Functional testing – Sprint 2</i>				
<i>System testing</i>				
<i>Regression testing</i>				
<i>Usability Testing</i>				
<i>Resolution of final defects and final build testing</i>				
<i>Deploy to Staging environment</i>				
<i>Performance testing</i>				
<i>Release to Production</i>				